

SECTION A

1 class vs Instance variable

In [13]:

```
class student:
    school_name="UPES"
    def __init__(self,name,age):
        self.name=name
        self.age=age

    def display(self):
        print(self.name, self.age)
        # print("Age= ",self.age)

print("School Name ", student.school_name)
s1=student("Aman",20)
s2=student("Riya",21)

print("Student1 : ",s1.display())
print("Student2 : ",s1.display())
# print("Student 1", (self.name) s1.(self.age))
```

```
School Name  UPES
Aman 20
Student1 :  None
Aman 20
Student2 :  None
```

Q2 class method vs Instance method

In [20]:

```
class Company:
    comapany= "ABC Ltd"

    def __init__(self,name,salary):
        self.name=name
        self.salary=salary

    def display(self):
        print("Employee Details")
        print(Company.comapany)
        print(self.name , self.salary)

    @classmethod
    def change_company(cls,new_name):
        Company.comapany=new_name

c=Company("Roahn",80256)
c.display()

Company.change_company("XYZ Company")
c.display()

# c.display("Rohan",25831)
```

```
Employee Details
ABC Ltd
Roahn 80256
Employee Details
XYZ Company
Roahn 80256
```

Q3 Static Method

In []:

```
class Mathhelper:
    @staticmethod
    def is_even(num):
        print(if num%2==0)
```

Section B

Q4 Private vs Public Member

In [29]:

```
class Bankaccount:
    bank_name="State Bank"

    def __init__(self, __amount):
        self.__amount=__amount

    # def deposit(self,amt):
    #     self.__amount+=amt
    def __add__(self,other,amt):
        self.__amount+=other.amt

    # def withdraw(self,amt):
    #     self.__amount-=amt
    def __sub__(self,other,amt):
        self.__amount-=other.amt

    def show_balance(self):
        print(self.__amount)
c=Bankaccount(500)
c.show_balance()
# c.deposit("300")
# c.withdraw("100")

c.show_balance()
```

500

500

Q5

In [38]:

```
class Student:
    """This is a docstring of the class """
    def __init__(self,name):
        self.name=name

print(Student.__doc__)
print(Student.__name__)
print(Student.__module__)
```

This is a docstring of the class

Student

__main__

Q6

In [35]:

```
class product:
    def __init__(self, name,price):
        self.name=name
        self.price=price

    def __str__(self):
        return f"{self.name} , {self.price}"
    def __add__(self,other):
        return f"{self.price+other.price}"

    def __eq__(self,other):
        return f"{self.price==other.price}"

p1 = product("Laptop", 50000)
p2 = product("Phone", 30000)
p3 = product("Tablet", 50000)

print(p1)
print(p1 + p2)
print(p1 == p3)
print(p1 == p2)
```

Laptop , 50000
80000
True
False